Arista Access Node BTM-ACC-19 SOP for Optical Transport Failure (SFP/XFP Degradation, Fiber Cuts, Rx/Tx Power Loss)

Node: BTM-ACC-19 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_7104

Description

Optical Transport Failures on Arista Access Nodes, specifically BTM-ACC-19, occur due to the degradation of SFP/XFP modules, fiber cuts, or a decline in receive/transmit power levels. This is primarily attributed to the inherent vulnerability of optical fibers to environmental stressors, including temperature fluctuations, mechanical stress, or contamination. Furthermore, the use of SFP/XFP modules, which facilitate high-speed data transfer through optical interfaces, introduces an additional layer of complexity. In the event of a failure, the role of the Arista Access Node is compromised, disrupting the flow of traffic and potentially impacting downstream services. This SOP outlines the diagnostic and resolution procedures necessary to rectify the issue, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- id: 1, description: Verify site location: BTM Layout Aggregation

- id: 2, description: Confirm Node ID: BTM-ACC-19

- id: 3, description: Check redundant link status to ensure alternate path availability

Potential Root Causes

Cause	Probability
SFP/XFP module degradation	High
Fiber cut or damage	Medium
Rx/Tx power level decline	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface Ethernet 1/1, Expected Output: Ethernet1/1 is up, line protocol is up (link is up), Description: This command checks the operational status of the interface, verifying its up/down state and link integrity.

Command: show sfp detail, Expected Output: SFP-10G-SR is not present., Description: This command retrieves detailed information about the SFP/XFP module, including its presence, status, and optical characteristics.

Command: show optical-monitor, Expected Output: RX Power: -20.00 dBm, Description: This command displays real-time optical monitoring data,
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including receive power levels, which can indicate potential issues
with the SFP/XFP module or fiber connection.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP module, Command: replace interface Ethernet 1/1 with 10G-SR, Description: This step involves replacing the faulty SFP/XFP module with a new one, ensuring optimal optical interface performance and network uptime.

Step 2: Step: Verify fiber connection, Command: show interface Ethernet 1/1, Description: After replacing the SFP/XFP module, verify the interface status to ensure the fiber connection is stable and functional.

Step 3: Step: Adjust Rx/Tx power levels, Command: adjust optical-monitor rx-power -10 dBm, Description: If the receive power level is too low, adjust the Rx/Tx power levels to ensure optimal optical interface performance and prevent further degradation.

Post-Check Verification

```
Command: show interface Ethernet 1/1, Expected Output: Ethernet1/1 is
up, line protocol is up (link is up), Description: Verify the
interface status to ensure the fiber connection is stable and
functional.

Command: show sfp detail, Expected Output: SFP-10G-SR is present and
operational., Description: Verify the SFP/XFP module status to ensure
it is present and operational.

Command: show optical-monitor, Expected Output: RX Power: -10.00 dBm,
Description: Verify the receive power level to ensure it is within
optimal ranges.
```

Roll-back Procedures

- Step: Revert SFP/XFP module change, Command: undo replace interface Ethernet 1/1 with 10G-SR, Description: If the fix fails, revert the SFP/XFP module replacement to its original state.

- Step: Re-establish fiber connection, Command: show interface Ethernet 1/1, Description: Re-establish the fiber connection and verify its status.

- Step: Re-adjust Rx/Tx power levels, Command: undo adjust optical-monitor rx-power -10 dBm, Description: Re-adjust the Rx/Tx power levels to their original settings.